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Chapter 13

Factorisation

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Exercise 13(A)

Question 1(i)

 $-7x^2 - 14y$ is equal to:

1. $-21x^2y$

2. $14x^2y$

3. $7(-x^2 - 2y)$



Chapter 19
Representing 3-D in 2-D

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4. $-7(x^2 + 2y)$

Answer

$$-7x^2 - 14y$$

$$= (-7 \times 1) x^2 + (-7 \times 2) y$$

$$= -7(x^2 + 2y)$$

Hence, option 4 is the correct option.

Question 1(ii)

$a(x - y) - b(x - y)^2$ is equal to:

1. $(x - y)^2 (a - b)$

2. $(x + y) (x - y) (a - b)$

3. $(x - y) (a - bx + by)$

4. $(x - y) (a - bx - by)$

Answer

$$a(x - y) - b(x - y)^2$$

$$= a(x - y) - b(x - y)(x - y)$$

$$= (x - y)(a - b(x - y))$$

$$= (x - y)(a - bx + by)$$

Hence, option 3 is the correct option.

Question 1(iii)

$a^2 + bc + ab + ac$ is equal to:

1. $(a + b)(a + c)$

2. $(a + b)(b + c)$

3. $(a + c)(a - b)$

4. $(a - b)(b + c)$

Answer

$$a^2 + bc + ab + ac$$

$$= (a \times a + ab) + (ac + bc)$$

$$= a(a + b) + c(a + b)$$

$$= (a + b)(a + c)$$

Hence, option 1 is the correct option.

Question 1(iv)

$1 - 2x - 2x^2 + 4x^3$ is equal to:

1. $(1 + 2x)(1 - 2x^2)$

2. $(1 + 2x)(1 + 2x^2)$

$$3. (1 - 2x) (1 - 2x^2)$$

$$4. (1 - 2x) (1 + 2x^2)$$

Answer

$$1 - 2x - 2x^2 + 4x^3$$

$$= 1 \times (1 - 2x) - (2x^2 - 2 \times 2x^2 \times x)$$

$$= 1 \times (1 - 2x) - 2x^2(1 - 2x)$$

$$= (1 - 2x)(1 - 2x^2)$$

Hence, option 3 is the correct option.

Question 1(v)

$a(x - y) - b(y - x)^2$ is equal to:

$$1. (x - y)(a - by + bx)$$

$$2. (y - x)(a - bx + by)$$

$$3. (x - y) (a - bx - by)$$

$$4. (x - y) (a - bx + by)$$

Answer

$$a(x - y) - b(y - x)^2$$

$$= a(x - y) - b(-(x - y))^2$$

$$= a(x - y) - b(x - y)^2$$

$$= (x - y)(a - b(x - y))$$

$$= (x - y)(a - bx + by)$$

Hence, option 4 is the correct option.

Question 2

$$17a^6b^8 - 34a^4b^6 + 51a^2b^4$$

Answer

$$17a^6b^8 - 34a^4b^6 + 51a^2b^4$$

$$= 17a^6b^8 - 17 \times 2a^4b^6 + 17 \times 3a^2b^4$$

$$= 17a^2b^4(a^4b^4 - 2a^2b^2 + 3)$$

$$\text{Hence, } 17a^6b^8 - 34a^4b^6 + 51a^2b^4 = 17a^2b^4(a^4b^4 - 2a^2b^2 + 3)$$

Question 3

$$3x^5y - 27x^4y^2 + 12x^3y^3$$

Answer

$$3x^5y - 27x^4y^2 + 12x^3y^3$$

$$= 3x^5y - 3 \times 9x^4y^2 + 3 \times 4x^3y^3$$

$$= 3x^3y(x^2 - 9xy + 4y^2)$$

$$\text{Hence, } 3x^5y - 27x^4y^2 + 12x^3y^3 = 3x^3y(x^2 - 9xy + 4y^2)$$

Question 4

$$x^2(a - b) - y^2(a - b) + z^2(a - b)$$

Answer

$$x^2(a - b) - y^2(a - b) + z^2(a - b)$$

$$= (a - b)(x^2 - y^2 + z^2)$$

$$\text{Hence, } x^2(a - b) - y^2(a - b) + z^2(a - b) = (a - b)(x^2 - y^2 + z^2)$$

Question 5

$$(x + y)(a + b) + (x - y)(a + b)$$

Answer

$$(x + y)(a + b) + (x - y)(a + b)$$

$$= (a + b)((x + y) + (x - y))$$

$$= (a + b)(x + y + x - y)$$

$$= (a + b)(2x)$$

$$= 2x(a + b)$$

$$\text{Hence, } (x + y)(a + b) + (x - y)(a + b) = 2x(a + b)$$

Question 6

$$2b(2a + b) - 3c(2a + b)$$

Answer

$$2b(2a + b) - 3c(2a + b)$$

$$= (2a + b)(2b - 3c)$$

$$\text{Hence, } 2b(2a + b) - 3c(2a + b) = (2a + b)(2b - 3c)$$

Question 7

$$12abc - 6a^2b^2c^2 + 3a^3b^3c^3$$

Answer

$$12abc - 6a^2b^2c^2 + 3a^3b^3c^3$$

$$= 3 \times 4abc - 3 \times 2a^2b^2c^2 + 3a^3b^3c^3$$

$$= 3abc(4 - 2abc + a^2b^2c^2)$$

$$\text{Hence, } 12abc - 6a^2b^2c^2 + 3a^3b^3c^3 = 3abc(4 - 2abc + a^2b^2c^2)$$

Question 8

$$4x(3x - 2y) - 2y(3x - 2y)$$

Answer

$$4x(3x - 2y) - 2y(3x - 2y)$$

$$= (3x - 2y)(4x - 2y)$$

$$= (3x - 2y)2(2x - y)$$

$$= 2(3x - 2y)(2x - y)$$

$$\text{Hence, } 4x(3x - 2y) - 2y(3x - 2y) = 2(3x - 2y)(2x - y)$$

Question 9

$$(a + 2b)(3a + b) - (a + b)(a + 2b) + (a + 2b)^2$$

Answer

$$(a + 2b)(3a + b) - (a + b)(a + 2b) + (a + 2b)^2$$

$$= (a + 2b) \left((3a + b) - (a + b) + (a + 2b) \right)$$

$$= (a + 2b) \left(3a + b - a - b + a + 2b \right)$$

$$= (a + 2b)(3a - a + a + b + 2b - b)$$

$$= (a + 2b)(3a + 2b)$$

$$\text{Hence, } (a + 2b)(3a + b) - (a + b)(a + 2b) + (a + 2b)^2 = (a + 2b)(3a + 2b)$$

Question 10

$$6xy(a^2 + b^2) + 8yz(a^2 + b^2) - 10xz(a^2 + b^2)$$

Answer

$$6xy(a^2 + b^2) + 8yz(a^2 + b^2) - 10xz(a^2 + b^2)$$

$$= (a^2 + b^2)(6xy + 8yz - 10xz)$$

$$= (a^2 + b^2)2(3xy + 4yz - 5xz)$$

$$= 2(a^2 + b^2)(3xy + 4yz - 5xz)$$

$$\text{Hence, } 6xy(a^2 + b^2) + 8yz(a^2 + b^2) - 10xz(a^2 + b^2) = 2(a^2 + b^2)(3xy + 4yz - 5xz)$$

Question 11

$$xy - ay - ax + a^2 + bx - ab$$

Answer

$$xy - ay - ax + a^2 + bx - ab$$

$$= y(x - a) - a(x - a) + b(x - a)$$

$$= (x - a)(y - a + b)$$

$$\text{Hence, } xy - ay - ax + a^2 + bx - ab = (x - a)(y - a + b)$$

Question 12

$$3x^5 - 6x^4 - 2x^3 + 4x^2 + x - 2$$

Answer

$$3x^5 - 6x^4 - 2x^3 + 4x^2 + x - 2$$

$$= 3x^4(x - 2) - 2x^2(x - 2) + 1(x - 2)$$

$$= (x - 2)(3x^4 - 2x^2 + 1)$$

$$\text{Hence, } 3x^5 - 6x^4 - 2x^3 + 4x^2 + x - 2 = (x - 2)(3x^4 - 2x^2 + 1)$$

Question 13

$$-x^2y - x + 3xy + 3$$

Answer

$$-x^2y - x + 3xy + 3$$

$$= -x(xy + 1) + 3(xy + 1)$$

$$= (xy + 1)(-x + 3)$$

$$= (xy + 1)(3 - x)$$

$$\text{Hence, } -x^2y - x + 3xy + 3 = (xy + 1)(3 - x)$$

Question 14

$$6a^2 - 3a^2b - bc^2 + 2c^2$$

Answer

$$6a^2 - 3a^2b - bc^2 + 2c^2$$

$$= 3a^2(2 - b) - c^2(b - 2)$$

$$= 3a^2(2 - b) + c^2(2 - b)$$

$$= (2 - b)(3a^2 + c^2)$$

$$\text{Hence, } 6a^2 - 3a^2b - bc^2 + 2c^2 = (2 - b)(3a^2 + c^2)$$

Question 15

$$3a^2b - 12a^2 - 9b + 36$$

Answer

$$3a^2b - 12a^2 - 9b + 36$$

$$= 3a^2(b - 4) - 9(b - 4)$$

$$= (b - 4)(3a^2 - 9)$$

$$= (b - 4)3(a^2 - 3)$$

$$= 3(b - 4)(a^2 - 3)$$

$$\text{Hence, } 3a^2b - 12a^2 - 9b + 36 = 3(b - 4)(a^2 - 3)$$

Question 16

$$x^2 - (a - 3)x - 3a$$

Answer

$$x^2 - (a - 3)x - 3a$$

$$= x^2 - ax + 3x - 3a$$

$$= x(x - a) + 3(x - a)$$

$$= (x - a)(x + 3)$$

$$\text{Hence, } x^2 - (a - 3)x - 3a = (x - a)(x + 3)$$

Question 17

$$ab^2 - (a - c) b - c$$

Answer

$$ab^2 - (a - c) b - c$$

$$= ab^2 - ab + bc - c$$

$$= ab(b - 1) + c(b - 1)$$

$$= (b - 1)(ab + c)$$

$$\text{Hence, } ab^2 - (a - c) b - c = (b - 1)(ab + c)$$

Question 18

$$(a^2 - b^2) c + (b^2 - c^2) a$$

Answer

$$(a^2 - b^2) c + (b^2 - c^2) a$$

$$= a^2c - c^2a + b^2a - b^2c$$

$$= ac(a - c) + b^2(a - c)$$

$$= (a - c)(ac + b^2)$$

$$\text{Hence, } (a^2 - b^2) c + (b^2 - c^2) a = (a - c)(ac + b^2)$$

Question 19

$$a^3 - a^2 - ab + a + b - 1$$

Answer

$$a^3 - a^2 - ab + a + b - 1$$

$$= a^3 - a^2 - ab + b + a - 1$$

$$= a^2(a - 1) - b(a - 1) + 1(a - 1)$$

$$= (a - 1)(a^2 - b + 1)$$

$$\text{Hence, } a^3 - a^2 - ab + a + b - 1 = (a - 1)(a^2 - b + 1)$$

Question 20

$$ab(c^2 + d^2) - a^2cd - b^2cd$$

Answer

$$ab(c^2 + d^2) - a^2cd - b^2cd$$

$$= abc^2 + abd^2 - a^2cd - b^2cd$$

$$= (abc^2 - a^2cd) + (abd^2 - b^2cd)$$

$$= ac(bc - ad) + bd(ad - bc)$$

$$= ac(bc - ad) - bd(bc - ad)$$

$$= (bc - ad)(ac - bd)$$

$$\text{Hence, } ab(c^2 + d^2) - a^2cd - b^2cd = (bc - ad)(ac - bd)$$

Question 21

$$2ab^2 - aby + 2cby - cy^2$$

Answer

$$2ab^2 - aby + 2cby - cy^2$$

$$= ab(2b - y) + cy(2b - y)$$

$$= (2b - y)(ab + cy)$$

$$\text{Hence, } 2ab^2 - aby + 2cby - cy^2 = (2b - y)(ab + cy)$$

Question 22

$$ax + 2bx + 3cx - 3a - 6b - 9c$$

Answer

$$ax + 2bx + 3cx - 3a - 6b - 9c$$

$$= x(a + 2b + 3c) - 3(a + 2b + 3c)$$

$$= (a + 2b + 3c)(x - 3)$$

$$\text{Hence, } ax + 2bx + 3cx - 3a - 6b - 9c = (a + 2b + 3c)(x - 3)$$

Question 23

$$2ab^2c - 2a + 3b^3c - 3b - 4b^2c^2 + 4c$$

Answer

$$2ab^2c - 2a + 3b^3c - 3b - 4b^2c^2 + 4c$$

$$= 2a(b^2c - 1) + 3b(b^2c - 1) - 4c(b^2c - 1)$$

$$= (b^2c - 1)(2a + 3b - 4c)$$

$$\text{Hence, } 2ab^2c - 2a + 3b^3c - 3b - 4b^2c^2 + 4c = (b^2c - 1)(2a + 3b - 4c)$$

Exercise 13(B)

Question 1(i)

$(2x + y)^2 - (2y + x)^2$ is equal to:

1. $3(x + y)(x - y)$
2. $2(x - y)(x + y)$
3. $2(y - x)(x + y)$
4. $(3x + y)(3x - y)$

Answer

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$(2x + y)^2 - (2y + x)^2$$

$$= ((2x + y) + (2y + x))((2x + y) - (2y + x))$$

$$= (2x + y + 2y + x)(2x + y - 2y - x)$$

$$= (2x + x + y + 2y)(2x - x + y - 2y)$$

$$= (3x + 3y)(x - y)$$

$$= 3(x + y)(x - y)$$

Hence, option 1 is the correct option.

Question 1(ii)

$49 - (x + 5)^2$ is equal to:

1. $(54 - x)(54 + x)$

2. $(2 - x)(12 + x)$

3. $48(x + 5)^2$

4. $48(x - 5)^2$

Answer

$$49 - (x + 5)^2$$

$$= 7^2 - (x + 5)^2$$

Using the formula

$$\begin{aligned} [\because (x^2 - y^2) &= (x + y)(x - y)] \\ &= ((7) + (x + 5))((7) - (x + 5)) \\ &= (7 + x + 5)(7 - x - 5) \\ &= (12 + x)(2 - x) \end{aligned}$$

Hence, option 2 is the correct option.

Question 1(iii)

$a^2 - 2ab + b^2 + a - b$ is equal to :

1. $(a - b)(a + b - 1)$
2. $(a - b)(a + b + 1)$
3. $(a + b)(a - b - 1)$
4. $(a - b)(a - b + 1)$

Answer

$$a^2 - 2ab + b^2 + a - b$$

Using the formula

$$\begin{aligned} [\because (x - y)^2 &= x^2 - 2xy + y^2] \\ &= (a^2 - 2ab + b^2) + a - b \end{aligned}$$

$$= (a - b)^2 + 1(a - b)$$

$$= (a - b)((a - b) + 1)$$

$$= (a - b)(a - b + 1)$$

Hence, option 4 is the correct option.

Question 1(iv)

$x^2 + y^2 - 2xy - 1$ is equal to :

1. $(x + y - 1)(x - y - 1)$

2. $(x + y + 1)(x - y - 1)$

3. $(x + y + 1)(x - y + 1)$

4. $(x - y + 1)(x - y - 1)$

Answer

$$x^2 - y^2 - 2xy - 1$$

Using the formula

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$= (x^2 + y^2 - 2xy) - 1$$

$$= (x - y)^2 - 1^2$$

Using the formula

$$\begin{aligned} [\because (x^2 + y^2) &= (x + y)(x - y)] \\ &= ((x - y) + (1))((x - y) - (1)) \\ &= (x - y + 1)(x - y - 1) \end{aligned}$$

Hence, option 4 is the correct option.

Question 1(v)

$a^2 + 2a + 1 - b^2 - x^2 + 2bx$ is equal to :

1. $(a + 1 - b + x)(a - 1 - b + x)$
2. $(a + 1 + b - x)(a + 1 - b + x)$
3. $(a - 1 + b - x)(a - 1 - b + x)$
4. $(a - 1 + bx)(a + 1 - bx)$

Answer

$$a^2 + 2a + 1 - b^2 - x^2 + 2bx$$

Using the formula

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

And

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

$$= (a^2 + 2a + 1) - (b^2 + x^2 - 2bx)$$

$$= (a + 1)^2 - (b - x)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((a + 1) + (b - x))((a + 1) - (b - x))$$

$$= (a + 1 + b - x)(a + 1 - b + x)$$

Hence, option 2 is the correct option.

Question 2

$$(a + 2b)^2 - a^2$$

Answer

$$(a + 2b)^2 - a^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((a + 2b) + a)((a + 2b) - a)$$

$$= (a + 2b + a)(a + 2b - a)$$

$$= (a + a + 2b)(a - a + 2b)$$

$$= (2a + 2b)(2b)$$

$$= 2(a + b)2b$$

$$= 4b(a + b)$$

$$\text{Hence, } (a + 2b)^2 - a^2 = 4b(a + b)$$

Question 3

$$(5a - 3b)^2 - 16b^2$$

Answer

$$(5a - 3b)^2 - 16b^2$$

$$= (5a - 3b)^2 - (4b)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((5a - 3b) + 4b)((5a - 3b) - 4b)$$

$$= (5a - 3b + 4b)(5a - 3b - 4b)$$

$$= (5a + 1b)(5a - 7b)$$

$$\text{Hence, } (5a - 3b)^2 - 16b^2 = (5a + b)(5a - 7b)$$

Question 4

$$a^4 - (a^2 - 3b^2)^2$$

Answer

$$a^4 - (a^2 - 3b^2)^2$$

$$= (a^2)^2 - (a^2 - 3b^2)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((a^2) + (a^2 - 3b^2))((a^2) - (a^2 - 3b^2))$$

$$= (a^2 + a^2 - 3b^2)(a^2 - a^2 + 3b^2)$$

$$= (2a^2 - 3b^2)(3b^2)$$

$$\text{Hence, } a^4 - (a^2 - 3b^2)^2 = 3b^2(2a^2 - 3b^2)$$

Question 5

$$(5a - 2b)^2 - (2a - b)^2$$

Answer

$$(5a - 2b)^2 - (2a - b)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((5a - 2b) + (2a - b))((5a - 2b) - (2a - b))$$

$$= (5a - 2b + 2a - b)(5a - 2b - 2a + b)$$

$$= (5a + 2a - 2b - b)(5a - 2a - 2b + b)$$

$$= (7a - 3b)(3a - b)$$

$$\text{Hence, } (5a - 2b)^2 - (2a - b)^2 = (7a - 3b)(3a - b)$$

Question 6

$$1 - 25(a + b)^2$$

Answer

$$1 - 25(a + b)^2$$

$$= 1^2 - (5(a + b))^2$$

$$= 1^2 - (5a + 5b)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((1) + (5a + 5b))((1) - (5a + 5b))$$

$$= (1 + 5a + 5b)(1 - 5a - 5b)$$

$$\text{Hence, } 1 - 25(a + b)^2 = (1 + 5a + 5b)(1 - 5a - 5b)$$

Question 7

$$4(2a + b)^2 - (a - b)^2$$

Answer

$$4(2a + b)^2 - (a - b)^2$$

$$= (2(2a + b))^2 - (a - b)^2$$

$$= (4a + 2b)^2 - (a - b)^2$$

Using the formula

$$\begin{aligned}
 [\because (x^2 - y^2) &= (x + y)(x - y)] \\
 &= ((4a + 2b) + (a - b))((4a + 2b) - (a - b)) \\
 &= (4a + 2b + a - b)(4a + 2b - a + b) \\
 &= (4a + a + 2b - b)(4a - a + 2b + b) \\
 &= (5a + b)(3a + 3b) \\
 &= (5a + b)3(a + b)
 \end{aligned}$$

$$\text{Hence, } 4(2a + b)^2 - (a - b)^2 = 3(5a + b)(a + b)$$

Question 8

$$25(2x + y)^2 - 16(x - y)^2$$

Answer

$$\begin{aligned}
 &25(2x + y)^2 - 16(x - y)^2 \\
 &= (5(2x + y))^2 - (4(x - y))^2 \\
 &= (10x + 5y)^2 - (4x - 4y)^2
 \end{aligned}$$

Using the formula

$$\begin{aligned}
 [\because (x^2 - y^2) &= (x + y)(x - y)] \\
 &= ((10x + 5y) + (4x - 4y))((10x + 5y) - (4x - 4y))
 \end{aligned}$$

$$= (10x + 5y + 4x - 4y)(10x + 5y - 4x + 4y)$$

$$= (10x + 4x + 5y - 4y)(10x - 4x + 5y + 4y)$$

$$= (14x + y)(6x + 9y)$$

$$= (14x + y)3(2x + 3y)$$

$$\text{Hence, } 25(2x + y)^2 - 16(x - y)^2 = 3(14x + y)(2x + 3y)$$

Question 9

$$\left(6\frac{2}{3}\right)^2 - \left(2\frac{1}{3}\right)^2$$

Answer

$$\begin{aligned} & \left(6\frac{2}{3}\right)^2 - \left(2\frac{1}{3}\right)^2 \\ &= \left(\frac{20}{3}\right)^2 - \left(\frac{7}{3}\right)^2 \end{aligned}$$

Using the formula

$$\begin{aligned} & [\because (x^2 - y^2) = (x + y)(x - y)] \\ &= \left(\frac{20}{3} + \frac{7}{3}\right)\left(\frac{20}{3} - \frac{7}{3}\right) \\ &= \left(\frac{(20 + 7)}{3}\right)\left(\frac{(20 - 7)}{3}\right) \\ &= \left(\frac{27}{3}\right)\left(\frac{13}{3}\right) \end{aligned}$$

$$= \left(\frac{27 \times 13}{3 \times 3} \right)$$

$$= \left(\frac{351}{9} \right)$$

$$= 39$$

$$\text{Hence, } \left(6\frac{2}{3}\right)^2 - \left(2\frac{1}{3}\right)^2 = 39$$

Question 10

$$(0.7)^2 - (0.3)^2$$

Answer

$$(0.7)^2 - (0.3)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (0.7 + 0.3)(0.7 - 0.3)$$

$$= (1.0) \times (0.4)$$

$$= 0.4$$

$$\text{Hence, } (0.7)^2 - (0.3)^2 = 0.4$$

Question 11

$$75(x + y)^2 - 48(x - y)^2$$

Answer

$$75(x + y)^2 - 48(x - y)^2$$

$$= 3 \times (25(x + y)^2 - 16(x - y)^2)$$

$$= 3 \times [(5(x + y))^2 - (4(x - y))^2]$$

$$= 3 \times [(5x + 5y)^2 - (4x - 4y)^2]$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= 3((5x + 5y) + (4x - 4y))((5x + 5y) - (4x - 4y))$$

$$= 3(5x + 5y + 4x - 4y)(5x + 5y - 4x + 4y)$$

$$= 3(5x + 4x + 5y - 4y)(5x - 4x + 5y + 4y)$$

$$= 3(9x + y)(x + 9y)$$

$$\text{Hence, } 75(x + y)^2 - 48(x - y)^2 = 3(9x + y)(x + 9y)$$

Question 12

$$a^2 + 4a + 4 - b^2$$

Answer

$$a^2 + 4a + 4 - b^2$$

Using the formula

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$= (a^2 + 4a + 4) - b^2$$

$$= (a + 2)^2 - b^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((a + 2) + b)((a + 2) - b)$$

$$= (a + 2 + b)(a + 2 - b)$$

$$\text{Hence, } a^2 + 4a + 4 - b^2 = (a + 2 + b)(a + 2 - b)$$

Question 13

$$a^2 - b^2 - 2b - 1$$

Answer

$$a^2 - b^2 - 2b - 1$$

Using the formula

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$= a^2 - (b^2 + 2b + 1)$$

$$= a^2 - (b + 1)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (a + (b + 1))(a - (b + 1))$$

$$= (a + b + 1)(a - b - 1)$$

$$\text{Hence, } a^2 - b^2 - 2b - 1 = (a + b + 1)(a - b - 1)$$

Question 14

$$x^2 + 6x + 9 - 4y^2$$

Answer

$$x^2 + 6x + 9 - 4y^2$$

Using the formula

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$= (x^2 + 6x + 9) - 4y^2$$

$$= (x + 3)^2 - (2y)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((x + 3) + (2y))((x + 3) - (2y))$$

$$= (x + 3 + 2y)(x + 3 - 2y)$$

$$\text{Hence, } x^2 + 6x + 9 - 4y^2 = (x + 3 + 2y)(x + 3 - 2y)$$

Exercise 13(C)

Question 1(i)

$x^2 - 9x - 10$ is equal to :

1. $(x - 10)(x + 1)$

2. $(x - 10)(x - 1)$

3. $(x + 10)(x - 1)$

4. $(x + 10)(x + 1)$

Answer

$$x^2 - 9x - 10$$

$$= x^2 - (10 - 1)x - 10$$

$$= x^2 - 10x + 1x - 10$$

$$= (x^2 - 10x) + (1x - 10)$$

$$= x(x - 10) + 1(x - 10)$$

$$= (x - 10)(x + 1)$$

Hence, option 1 is the correct option.

Question 1(ii)

$x^2 - 23x + 42$ is equal to :

1. $(x - 21)(x + 2)$

2. $(x - 21)(x - 2)$

3. $(x + 21)(x + 2)$

4. $(x + 21)(x - 2)$

Answer

$$x^2 - 23x + 42$$

$$= x^2 - (21 + 2)x + 42$$

$$= x^2 - 21x - 2x + 42$$

$$= (x^2 - 21x) - 2(2x - 42)$$

$$= x(x - 21) - 2(x - 21)$$

$$= (x - 21)(x - 2)$$

Hence, option 2 is the correct option.

Question 1(iii)

$(4x^2 - 4x + 1) \div (2x - 1)$ is equal to :

1. $2x + 1$
2. $2x - 1$
3. $2^x - 1$
4. none of these

Answer

$$(4x^2 - 4x + 1) \div (2x - 1)$$

$$= \frac{(4x^2 - 4x + 1)}{(2x - 1)}$$

Using the formula,

$$[\because (x - y)^2 = x^2 + y^2 - 2xy]$$

$$= \frac{((2x)^2 - 2 \times 2x \times 1 + 1)}{(2x - 1)}$$

$$= \frac{(2x - 1)^2}{(2x - 1)}$$

$$= \frac{(2x - 1)(2x - 1)}{(2x - 1)}$$

$$= \frac{(2x - 1)\cancel{(2x - 1)}}{\cancel{(2x - 1)}}$$

$$= 2x - 1$$

Hence, option 2 is the correct option.

Question 1(iv)

$(x + y)^2 - 3(x + y) - 4$ is equal to :

1. $(x + y + 4)(x + y - 1)$

2. $(x + y + 4)(x + y + 1)$

3. $(x + y - 4)(x + y + 1)$

4. $(x + y - 4)(x + y - 1)$

Answer

$$(x + y)^2 - 3(x + y) - 4$$

$$= (x + y)^2 + (-4 + 1)(x + y) - 4$$

$$= (x + y)^2 + (-4 + 1)(x + y) - 4$$

$$= (x + y)^2 - 4(x + y) + 1(x + y) - 4$$

$$= [(x + y)^2 - 4(x + y)] + [1(x + y) - 4]$$

$$= (x + y)[(x + y) - 4] + 1[(x + y) - 4]$$

$$= [(x + y) + 1][(x + y) - 4]$$

$$= [x + y + 1][x + y - 4]$$

Hence, option 3 is the correct option.

Question 1(v)

$60 + 11x - x^2$ is equal to :

1. $(4 + x)(15 - x)$

2. $(4 - x)(15 - x)$

3. $(4 - x)(15 + x)$

4. $(4 + x)(15 + x)$

Answer

$$60 + 11x - x^2$$

$$= 60 + (15 - 4)x - x^2$$

$$= 60 + 15x - 4x - x^2$$

$$= 15(4 + x) - x(4 + x)$$

$$= (4 + x)(15 - x)$$

Hence, option 1 is the correct option.

Question 2

$$a^2 + 5a + 6$$

Answer

$$a^2 + 5a + 6$$

$$= a^2 + (2 + 3)a + 6$$

$$= a^2 + 2a + 3a + 6$$

$$= (a^2 + 2a) + (3a + 6)$$

$$= a(a + 2) + 3(a + 2)$$

$$= (a + 2)(a + 3)$$

$$\text{Hence, } a^2 + 5a + 6 = (a + 2)(a + 3)$$

Question 3

$$a^2 - 5a + 6$$

Answer

$$a^2 - 5a + 6$$

$$= a^2 + (-2 - 3)a + 6$$

$$= a^2 + -2a - 3a + 6$$

$$= (a^2 - 2a) - (3a + 6)$$

$$= a(a - 2) - 3(a - 2)$$

$$= (a - 2)(a - 3)$$

$$\text{Hence, } a^2 - 5a + 6 = (a - 2)(a - 3)$$

Question 4

$$a^2 + 5a - 6$$

Answer

$$a^2 + 5a - 6$$

$$= a^2 + (6 - 1)a - 6$$

$$= a^2 + 6a - 1a - 6$$

$$= (a^2 + 6a) - (1a + 6)$$

$$= a(a + 6) - 1(a + 6)$$

$$= (a + 6)(a - 1)$$

$$\text{Hence, } a^2 + 5a - 6 = (a + 6)(a - 1)$$

Question 5

$$x^2 + 5xy + 4y^2$$

Answer

$$x^2 + 5xy + 4y^2$$

$$= x^2 + (4 + 1)xy + 4y^2$$

$$= x^2 + 4xy + 1xy + 4y^2$$

$$= (x^2 + 4xy) + (1xy + 4y^2)$$

$$= x(x + 4y) + y(x + 4y)$$

$$= (x + 4y)(x + y)$$

$$\text{Hence, } x^2 + 5xy + 4y^2 = (x + 4y)(x + y)$$

Question 6

$$a^2 - 3a - 40$$

Answer

$$a^2 - 3a - 40$$

$$= a^2 + (-8 + 5)a - 40$$

$$= a^2 - 8a + 5a - 40$$

$$= (a^2 - 8a) + (5a - 40)$$

$$= a(a - 8) + 5(a - 8)$$

$$= (a - 8)(a + 5)$$

$$\text{Hence, } a^2 - 3a - 40 = (a - 8)(a + 5)$$

Question 7

$$x^2 - x - 72$$

Answer

$$x^2 - x - 72$$

$$= x^2 + (-9 + 8)x - 72$$

$$= x^2 - 9x + 8x - 72$$

$$= x(x - 9) + 8(x - 9)$$

$$= (x - 9)(x + 8)$$

$$\text{Hence, } x^2 - x - 72 = (x - 9)(x + 8)$$

Question 8

$$3a^2 - 5a + 2$$

Answer

$$3a^2 - 5a + 2$$

$$= 3a^2 - (3 + 2)a + 2$$

$$= 3a^2 - 3a - 2a + 2$$

$$= (3a^2 - 3a) - (2a - 2)$$

$$= 3a(a - 1) - 2(a - 1)$$

$$= (a - 1)(3a - 2)$$

$$\text{Hence, } 3a^2 - 5a + 2 = (a - 1)(3a - 2)$$

Question 9

$$2a^2 - 17ab + 26b^2$$

Answer

$$\begin{aligned} & 2a^2 - 17ab + 26b^2 \\ &= 2a^2 - (13 + 4)ab + 26b^2 \\ &= 2a^2 - 13ab - 4ab + 26b^2 \\ &= (2a^2 - 13ab) - (4ab - 26b^2) \\ &= a(2a - 13b) - 2b(2a - 13b) \\ &= (2a - 13b)(a - 2b) \end{aligned}$$

Hence, $2a^2 - 17ab + 26b^2 = (2a - 13b)(a - 2b)$

Question 10

$$2x^2 + xy - 6y^2$$

Answer

$$\begin{aligned} & 2x^2 + xy - 6y^2 \\ &= 2x^2 + (4 - 3)xy - 6y^2 \\ &= 2x^2 + 4xy - 3xy - 6y^2 \\ &= (2x^2 + 4xy) - (3xy + 6y^2) \\ &= 2x(x + 2y) - 3y(x + 2y) \\ &= (x + 2y)(2x - 3y) \end{aligned}$$

$$\text{Hence, } 2x^2 + xy - 6y^2 = (x + 2y)(2x - 3y)$$

Question 11

$$4c^2 + 3c - 10$$

Answer

$$4c^2 + 3c - 10$$

$$= 4c^2 + (8 - 5)c - 10$$

$$= 4c^2 + 8c - 5c - 10$$

$$= (4c^2 + 8c) - (5c + 10)$$

$$= 4c(c + 2) - 5(c + 2)$$

$$= (c + 2)(4c - 5)$$

$$\text{Hence, } 4c^2 + 3c - 10 = (c + 2)(4c - 5)$$

Question 12

$$14x^2 + x - 3$$

Answer

$$14x^2 + x - 3$$

$$= 14x^2 + (7 - 6)x - 3$$

$$= 14x^2 + 7x - 6x - 3$$

$$= 7x(2x + 1) - 3(2x + 1)$$

$$= (2x + 1)(7x - 3)$$

$$\text{Hence, } 14x^2 + x - 3 = (2x + 1)(7x - 3)$$

Question 13

$$6 + 7b - 3b^2$$

Answer

$$6 + 7b - 3b^2$$

$$= 6 + (9 - 2)b - 3b^2$$

$$= 6 + 9b - 2b - 3b^2$$

$$= (6 + 9b) - (2b + 3b^2)$$

$$= 3(2 + 3b) - b(2 + 3b)$$

$$= (2 + 3b)(3 - b)$$

$$\text{Hence, } 6 + 7b - 3b^2 = (2 + 3b)(3 - b)$$

Question 14

$$5 + 7x - 6x^2$$

Answer

$$5 + 7x - 6x^2$$

$$= 5 + (10 - 3)x - 6x^2$$

$$= 5 + 10x - 3x - 6x^2$$

$$= (5 + 10x) - (3x + 6x^2)$$

$$= 5(1 + 2x) - 3x(1 + 2x)$$

$$= (1 + 2x)(5 - 3x)$$

$$\text{Hence, } 5 + 7x - 6x^2 = (1 + 2x)(5 - 3x)$$

Question 15

$$4 + y - 14y^2$$

Answer

$$4 + y - 14y^2$$

$$= 4 + (8 - 7)y - 14y^2$$

$$= 4 + 8y - 7y - 14y^2$$

$$= (4 + 8y) - (7y + 14y^2)$$

$$= 4(1 + 2y) - 7y(1 + 2y)$$

$$= (1 + 2y)(4 - 7y)$$

$$\text{Hence, } 4 + y - 14y^2 = (1 + 2y)(4 - 7y)$$

Question 16

$$5 + 3a - 14a^2$$

Answer

$$5 + 3a - 14a^2$$

$$= 5 + (10 - 7)a - 14a^2$$

$$= 5 + 10a - 7a - 14a^2$$

$$= (5 + 10a) - (7a + 14a^2)$$

$$= 5(1 + 2a) - 7a(1 + 2a)$$

$$= (1 + 2a)(5 - 7a)$$

$$\text{Hence, } 5 + 3a - 14a^2 = (1 + 2a)(5 - 7a)$$

Question 17

$$(2a + b)^2 + 5(2a + b) + 6$$

Answer

$$(2a + b)^2 + 5(2a + b) + 6$$

$$= (2a + b)^2 + (2 + 3)(2a + b) + 6$$

$$= (2a + b)^2 + 2(2a + b) + 3(2a + b) + 6$$

$$= [(2a + b)^2 + 2(2a + b)] + [3(2a + b) + 6]$$

$$= (2a + b)[(2a + b) + 2] + 3[(2a + b) + 2]$$

$$= [(2a + b) + 2][(2a + b) + 3]$$

$$= [2a + b + 2][2a + b + 3]$$

$$\text{Hence, } (2a + b)^2 + 5(2a + b) + 6 = (2a + b + 2)(2a + b + 3)$$

Question 18

$$1 - (2x + 3y) - 6(2x + 3y)^2$$

Answer

$$1 - (2x + 3y) - 6(2x + 3y)^2$$

$$= 1 - (3 - 2)(2x + 3y) - 6(2x + 3y)^2$$

$$= 1 - 3(2x + 3y) + 2(2x + 3y) - 6(2x + 3y)^2$$

$$= 1[1 - 3(2x + 3y)] + 2(2x + 3y)[1 - 3(2x + 3y)]$$

$$= [1 - 3(2x + 3y)][1 + 2(2x + 3y)]$$

$$= [1 - 6x - 9y][1 + 4x + 6y]$$

$$\text{Hence, } 1 - (2x + 3y) - 6(2x + 3y)^2 = [1 - 6x - 9y][1 + 4x + 6y]$$

Question 19

$$(x - 2y)^2 - 12(x - 2y) + 32$$

Answer

$$(x - 2y)^2 - 12(x - 2y) + 32$$

$$= (x - 2y)^2 - (8 + 4)(x - 2y) + 32$$

$$= (x - 2y)^2 - 8(x - 2y) - 4(x - 2y) + 32$$

$$= [(x - 2y)^2 - 8(x - 2y)] - [4(x - 2y) - 32]$$

$$= (x - 2y)[(x - 2y) - 8] - 4[(x - 2y) - 8]$$

$$= [(x - 2y) - 8][(x - 2y) - 4]$$

$$= [x - 2y - 8][x - 2y - 4]$$

$$\text{Hence, } (x - 2y)^2 - 12(x - 2y) + 32 = [x - 2y - 8][x - 2y - 4]$$

Question 20

$$8 + 6(a + b) - 5(a + b)^2$$

Answer

$$8 + 6(a + b) - 5(a + b)^2$$

$$= 8 + (10 - 4)(a + b) - 5(a + b)^2$$

$$= 8 + 10(a + b) - 4(a + b) - 5(a + b)^2$$

$$= [8 + 10(a + b)] - [4(a + b) + 5(a + b)^2]$$

$$= 2[4 + 5(a + b)] - (a + b)[4 + 5(a + b)]$$

$$= [4 + 5(a + b)][2 - (a + b)]$$

$$= [4 + 5a + 5b][2 - a - b]$$

$$\text{Hence, } 8 + 6(a + b) - 5(a + b)^2 = [4 + 5a + 5b][2 - a - b]$$

Question 21

$$2(x + 2y)^2 - 5(x + 2y) + 2$$

Answer

$$2(x + 2y)^2 - 5(x + 2y) + 2$$

$$= 2(x + 2y)^2 - (4 + 1)(x + 2y) + 2$$

$$= 2(x + 2y)^2 - 4(x + 2y) - 1(x + 2y) + 2$$

$$= [2(x + 2y)^2 - 4(x + 2y)] - [1(x + 2y) - 2]$$

$$= 2(x + 2y)[(x + 2y) - 2] - 1[(x + 2y) - 2]$$

$$= [(x + 2y) - 2][2(x + 2y) - 1]$$

$$= [x + 2y - 2][2x + 4y - 1]$$

$$\text{Hence, } 2(x + 2y)^2 - 5(x + 2y) + 2 = [x + 2y - 2][2x + 4y - 1]$$

Question 22(i)

Find whether the trinomial is a perfect square or not :

$$x^2 + 14x + 49$$

Answer

$$x^2 + 14x + 49$$

Using the formula

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

$$= x^2 + 2 \times 7 \times x + (7)^2$$

$$= (x + 7)^2$$

Hence, $x^2 + 14x + 49$ is a perfect square.

Question 22(ii)

Find whether the trinomial is a perfect square or not :

$$a^2 - 10a + 25$$

Answer

$$a^2 - 10a + 25$$

Using the formula

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$= a^2 - 2 \times a \times 5 + (5)^2$$

$$= (a - 5)^2$$

Hence, $a^2 - 10a + 25$ is a perfect square.

Question 22(iii)

Find whether the trinomial is a perfect square or not :

$$4x^2 + 4x + 1$$

Answer

$$4x^2 + 4x + 1$$

Using the formula

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

$$= (2x)^2 + 2 \times 2x \times 1 + (1)^2$$

$$= (2x + 1)^2$$

Hence, $4x^2 + 4x + 1$ is a perfect square.

Question 22(iv)

Find whether the trinomial is a perfect square or not :

$$9b^2 + 12b + 16$$

Answer

$$9b^2 + 12b + 16$$

Using the formula

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

Since, $(9b^2 + 12b + 16)$ cannot be expressed as $a^2 + 2ab + b^2$.

Hence, $9b^2 + 12b + 16$ is not a perfect square.

Question 22(v)

Find whether the trinomial is a perfect square or not :

$$16x^2 - 16xy + y^2$$

Answer

$$16x^2 - 16xy + y^2$$

Using the formula

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

Since, $(16x^2 - 16xy + y^2)$ cannot be expressed as $a^2 - 2ab + b^2$.

Hence, $16x^2 - 16xy + y^2$ is not a perfect square.

Question 22(vi)

Find whether the trinomial is a perfect square or not :

$$x^2 - 4x + 16$$

Answer

$$x^2 - 4x + 16$$

Using the formula

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

Since, $(x^2 - 4x + 16)$ cannot be expressed as $a^2 - 2ab + b^2$.

Hence, $x^2 - 4x + 16$ is not a perfect square.

Exercise 13(D)

Question 1(i)

$x^3 - 4x$ is equal to:

1. $x(x + 4)(x - 4)$

2. $x(x + 2)(x - 2)$

3. $(x + 4)(x - 4)$

4. $(x + 2)(x - 2)$

Answer

$$x^3 - 4x$$

$$= x(x^2 - 4)$$

$$= x ((x)^2 - (2)^2)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= x (x + 2)(x - 2)$$

Hence, option 2 is the correct option.

Question 1(ii)

$x^4 - y^4 + x^2 - y^2$ is equal to:

1. $(x + y + 1)(x + y - 1)(x^2 + y^2)$

2. $(x + y)(x - y)(x^2 + y^2 - 1)$

3. $(x + y)(x - y)(x^2 + y^2 + 1)$

4. none of these

Answer

$$x^4 - y^4 + x^2 - y^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (x^4 - y^4) + (x^2 - y^2)$$

$$= (x^2 - y^2)(x^2 + y^2) + (x - y)(x + y)$$

$$= (x - y)(x + y)(x^2 + y^2) + (x - y)(x + y)$$

$$= (x - y)(x + y)((x^2 + y^2) + 1)$$

$$= (x - y)(x + y)(x^2 + y^2 + 1)$$

Hence, option 3 is the correct option.

Question 1(iii)

$x^3 - x^2 + ax + x - a - 1$ is equal to:

1. $(x - 1)(x^2 + a - 1)$

2. $(x - 1)(x^2 + a + 1)$

3. $(x - 1)(x^2 - a + 1)$

4. $(x - 1)(x^2 - a - 1)$

Answer

$$x^3 - x^2 + ax + x - a - 1$$

$$= (x^3 - x^2) + (ax - a) + (x - 1)$$

$$= x^2(x - 1) + a(x - 1) + 1(x - 1)$$

$$= (x - 1)(x^2 + a + 1)$$

Hence, option 2 is the correct option.

Question 1(iv)

$8x^3 - 18x$ is equal to:

1. $x(2x + 3)(2x - 3)$

2. $2x(3 - 2x)(3 + 2x)$

3. $2x(2x + 3)(2x - 3)$

4. $x(4x + 6y)(4x - 6y)$

Answer

$$8x^3 - 18x$$

$$= 2x(4x^2 - 9)$$

$$= 2x((2x)^2 - (3)^2)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= 2x(2x - 3)(2x + 3)$$

Hence, option 3 is the correct option.

Question 1(v)

$x^2 - (a - b)x - ab$ is equal to:

1. $(x - a)(x - b)$

2. $(x + a)(x - b)$

$$3. (x - a)(x + b)$$

$$4. (x + a)(x + b)$$

Answer

$$x^2 - (a - b)x - ab$$

$$= x^2 - ax + bx - ab$$

$$= (x^2 - ax) + (bx - ab)$$

$$= x(x - a) + b(x - a)$$

$$= (x - a)(x + b)$$

Hence, option 3 is the correct option.

Question 2

$$8x^2y - 18y^3$$

Answer

$$8x^2y - 18y^3$$

$$= 2y(4x^2 - 9y^2)$$

$$= 2y((2x)^2 - (3y)^2)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= 2y (2x - 3y)(2x + 3y)$$

$$\text{Hence, } 8x^2y - 18y^3 = 2y (2x - 3y)(2x + 3y)$$

Question 3

$$25x^3 - x$$

Answer

$$25x^3 - x$$

$$= x (25x^2 - 1)$$

$$= x ((5x)^2 - (1)^2)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= x (5x + 1)(5x - 1)$$

$$\text{Hence, } 25x^3 - x = x (5x + 1)(5x - 1)$$

Question 4

$$16x^4 - 81y^4$$

Answer

$$16x^4 - 81y^4$$

$$= (4x^2)^2 - (9y^2)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (4x^2 + 9y^2)(4x^2 - 9y^2)$$

$$= (4x^2 + 9y^2)((2x)^2 - (3y)^2)$$

$$= (4x^2 + 9y^2)(2x - 3y)(2x + 3y)$$

$$\text{Hence, } 16x^4 - 81y^4 = (4x^2 + 9y^2)(2x - 3y)(2x + 3y)$$

Question 5

$$x^2 - y^2 - 3x - 3y$$

Answer

$$x^2 - y^2 - 3x - 3y$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (x^2 - y^2) - (3x + 3y)$$

$$= (x - y)(x + y) - 3(x + y)$$

$$= (x + y)((x - y) - 3)$$

$$= (x + y)(x - y - 3)$$

$$\text{Hence, } x^2 - y^2 - 3x - 3y = (x + y)(x - y - 3)$$

Question 6

$$x^2 - y^2 - 2x + 2y$$

Answer

$$x^2 - y^2 - 2x + 2y$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (x^2 - y^2) - (2x - 2y)$$

$$= (x - y)(x + y) - 2(x - y)$$

$$= (x - y)((x + y) - 2)$$

$$= (x - y)(x + y - 2)$$

$$\text{Hence, } x^2 - y^2 - 2x + 2y = (x - y)(x + y - 2)$$

Question 7

$$3x^2 + 15x - 72$$

Answer

$$3x^2 + 15x - 72$$

$$= 3(x^2 + 5x - 24)$$

$$= 3(x^2 + (8 - 3)x - 24)$$

$$= 3(x^2 + 8x - 3x - 24)$$

$$= 3[(x^2 + 8x) - (3x + 24)]$$

$$= 3[x(x + 8) - 3(x + 8)]$$

$$= 3(x + 8)(x - 3)$$

$$\text{Hence, } 3x^2 + 15x - 72 = 3(x + 8)(x - 3)$$

Question 8

$$2a^2 - 8a - 64$$

Answer

$$2a^2 - 8a - 64$$

$$= 2(a^2 - 4a - 32)$$

$$= 2(a^2 - (8 - 4)a - 32)$$

$$= 2(a^2 - 8a + 4a - 32)$$

$$= 2[(a^2 - 8a) + (4a - 32)]$$

$$= 2[a(a - 8) + 4(a - 8)]$$

$$= 2(a - 8)(a + 4)$$

$$\text{Hence, } 2a^2 - 8a - 64 = 2(a - 8)(a + 4)$$

Question 9

$$3x^2y + 11xy + 6y$$

Answer

$$3x^2y + 11xy + 6y$$

$$= 3x^2y + (9 + 2)xy + 6y$$

$$= 3x^2y + 9xy + 2xy + 6y$$

$$= (3x^2y + 9xy) + (2xy + 6y)$$

$$= 3xy(x + 3) + 2y(x + 3)$$

$$= (x + 3)(3xy + 2y)$$

$$= (x + 3)y(3x + 2)$$

$$\text{Hence, } 3x^2y + 11xy + 6y = y(x + 3)(3x + 2)$$

Question 10

$$5ap^2 + 11ap + 2a$$

Answer

$$5ap^2 + 11ap + 2a$$

$$= a(5p^2 + 11p + 2)$$

$$= a(5p^2 + (10 + 1)p + 2)$$

$$= a(5p^2 + 10p + 1p + 2)$$

$$= a[(5p^2 + 10p) + (1p + 2)]$$

$$= a[5p(p + 2) + 1(p + 2)]$$

$$= a(p + 2)(5p + 1)$$

$$\text{Hence, } 5ap^2 + 11ap + 2a = a(p + 2)(5p + 1)$$

Question 11

$$a^2 + 2ab + b^2 - c^2$$

Answer

$$a^2 + 2ab + b^2 - c^2$$

Using the formula

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$= (a^2 + 2ab + b^2) - c^2$$

$$= ((a)^2 + 2 \times a \times b + (b)^2) - c^2$$

$$= (a + b)^2 - (c)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((a + b) + c)((a + b) - c)$$

$$= (a + b + c)(a + b - c)$$

$$\text{Hence, } a^2 + 2ab + b^2 - c^2 = (a + b + c)(a + b - c)$$

Question 12

$$x^2 + 6xy + 9y^2 + x + 3y$$

Answer

$$x^2 + 6xy + 9y^2 + x + 3y$$

Using the formula

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$= (x^2 + 6xy + 9y^2) + x + 3y$$

$$= ((x)^2 + 2 \times x \times 3y + (3y)^2) + x + 3y$$

$$= (x + 3y)^2 + (x + 3y)$$

$$= (x + 3y)((x + 3y) + 1)$$

$$= (x + 3y)(x + 3y + 1)$$

$$\text{Hence, } x^2 + 6xy + 9y^2 + x + 3y = (x + 3y)(x + 3y + 1)$$

Question 13

$$4a^2 - 12ab + 9b^2 + 4a - 6b$$

Answer

$$4a^2 - 12ab + 9b^2 + 4a - 6b$$

Using the formula

$$[\because (x - y)^2 = x^2 + y^2 - 2xy]$$

$$= (4a^2 - 12ab + 9b^2) + 4a - 6b$$

$$= ((2a)^2 - 2 \times 2a \times 3b + (3b)^2) + 4a - 6b$$

$$= (2a - 3b)^2 + (4a - 6b)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (2a - 3b)^2 + 2(2a - 3b)$$

$$= (2a - 3b)((2a - 3b) + 2)$$

$$= (2a - 3b)(2a - 3b + 2)$$

$$\text{Hence, } 4a^2 - 12ab + 9b^2 + 4a - 6b = (2a - 3b)(2a - 3b + 2)$$

Question 14

$$2a^2b^2 - 98b^4$$

Answer

$$2a^2b^2 - 98b^4$$

$$= 2b^2(a^2 - 49b^2)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= 2b^2((a)^2 - (7b)^2)$$

$$= 2b^2(a - 7b)(a + 7b)$$

$$\text{Hence, } 2a^2b^2 - 98b^4 = 2b^2(a - 7b)(a + 7b)$$

Question 15

$$a^2 - 16b^2 - 2a - 8b$$

Answer

$$a^2 - 16b^2 - 2a - 8b$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (a^2 - 16b^2) - (2a + 8b)$$

$$= (a - 4b)(a + 4b) - 2(a + 4b)$$

$$= (a + 4b)((a - 4b) - 2)$$

$$= (a + 4b)(a - 4b - 2)$$

$$\text{Hence, } a^2 - 16b^2 - 2a - 8b = (a + 4b)(a - 4b - 2)$$

Test Yourself

Question 1(i)

$(a + b)^2 - 4ab$ is equal to:

1. $(a + b + 2ab)(a + b - 2ab)$

2. $(a + b)(a - b)$

3. $(a + b)(a + b)$

4. $(a - b)(a - b)$

Answer

$$(a + b)^2 - 4ab$$

Using the formula

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$= a^2 + b^2 + 2ab - 4ab$$

$$= a^2 + b^2 + (2ab - 4ab)$$

$$= a^2 + b^2 - 2ab$$

Using the formula

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$= (a)^2 + (b)^2 - 2 \times a \times b$$

$$= (a - b)^2$$

$$= (a - b)(a - b)$$

Hence, option 4 is the correct option.

Question 1(ii)

$a^4 + 4a^2 - 32$ is equal to:

1. $(a^2 + 8)(a + 2)(a + 2)$

2. $(a^2 - 8)(a - 2)(a + 2)$

3. $(a^2 + 8)(a^2 + 4)$

4. $(a^2 + 8)(a + 2)(a - 2)$

Answer

$$a^4 + 4a^2 - 32$$

$$= a^4 + (8 - 4)a^2 - 32$$

$$= a^4 + 8a^2 - 4a^2 - 32$$

$$= (a^4 + 8a^2) - (4a^2 + 32)$$

$$= a^2(a^2 + 8) - 4(a^2 + 8)$$

$$= (a^2 + 8)(a^2 - 4)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (a^2 + 8)(a - 2)(a + 2)$$

Hence, option 4 is the correct option.

Question 1(iii)

$36 - 60y + 25y^2$ is equal to :

1. $(3 + 5y)(3 + 5y)$

2. $(3 - 5y)(6 - 5y)$

3. $(3 + 4y)(3 - 4y)$

4. none of these

Answer

$$36 - 60y + 25y^2$$

Using the formula

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$= (6)^2 - 2 \times 6 \times 5y + (5y)^2$$

$$= (6 - 5y)^2$$

$$= (6 - 5y)(6 - 5y)$$

Hence, option 4 is the correct option.

Question 1(iv)

$(x - 2y)^2 - 3x + 6y$ is equal to :

1. $(x - 3y)(x + 2y)$

2. $(x - 2y)(x - 2y + 3)$

3. $(x + 2y - 3)(x + 2y)$

4. $(x - 2y)(x - 2y - 3)$

Answer

$$(x - 2y)^2 - 3x + 6y$$

$$= (x - 2y)^2 - 3(x - 2y)$$

$$= (x - 2y)((x - 2y) - 3)$$

$$= (x - 2y)(x - 2y - 3)$$

Hence, option 4 is the correct option.

Question 1(v)

$a(x - y)^2 - by + bx$ is equal to :

1. $(x - y)(ax + by + b)$

2. $(x - y)(ax + by - b)$

3. $(x - y)(x + y + a - b)$

4. $(x - y)(ax - ay + b)$

Answer

$$a(x - y)^2 - by + bx$$

$$= a(x - y)^2 - (by - bx)$$

$$= a(x - y)^2 - b(y - x)$$

$$= a(x - y)^2 + b(x - y)$$

$$= (x - y)(a(x - y) + b)$$

$$= (x - y)(ax - ay + b)$$

Hence, option 4 is the correct option.

Question 1(vi)

Statement 1: The product of two binomials is not necessarily a trinomial.

Statement 2: The difference of the square of two terms = The sum of the same two terms x their difference.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

According to statement 1 :

The product of two binomials is a trinomial.

For example; $(x + 2)(x + 3) = x(x + 3) + 2(x + 3)$

$$= x^2 + 3x + 2x + 6$$

$$= x^2 + 5x + 6$$

Here, $(x + 2)$ and $(x + 3)$ are two binomials and their product $x^2 + 5x + 6$ is a trinomial.

$$\text{But, } (x + 2)(x - 2) = x^2 - 2^2$$

$$= x^2 - 4$$

Here, $(x + 2)$ and $(x - 2)$ are two binomials but $x^2 - 4$ is not a trinomial.

So, we can say that the product of two binomials is not necessarily a trinomial.

So, statement 1 is true.

The difference of the square of two terms = the sum of the same two terms x their difference.

As,

$$a^2 - b^2 = (a - b)(a + b)$$

$$\text{L.H.S.} = a^2 - b^2$$

$$\text{R.H.S.} = (a - b) \times (a + b)$$

Therefore, L.H.S. = R.H.S.

$\Rightarrow$ The difference of the square of two terms = the sum of the same two terms x their difference.

So, statement 2 is also true.

Hence, option 1 is the correct option.

Question 1(vii)

Assertion (A) : $25x^2 - 5x + 1$ is a perfect square trinomial.

Reason (R) : Any trinomial which can be expressed as $x^2 + y^2 + 2xy$ or $x^2 + y^2 - 2xy$ is a perfect square trinomial.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

$$(5x - 1)^2 = (5x)^2 - 2.5x.1 + 1^2$$

$$= 25x^2 - 10x + 1^2$$

Thus, $25x^2 - 5x + 1$ is not a perfect square trinomial.

So, assertion (A) is false.

$$(x - y)^2 = x^2 - 2.x.y + y^2$$

$$= x^2 - 2xy + y^2$$

$$\text{And, } (x + y)^2 = x^2 + 2.x.y + y^2$$

$$= x^2 + 2xy + y^2$$

Thus, $x^2 + y^2 + 2xy$ or $x^2 + y^2 - 2xy$ is a perfect square trinomial.

So, reason (R) is true.

Hence, option 4 is the correct option.

Question 1(viii)

$$\text{Assertion (A) : } x^2 + 7x + 12$$

$$= x^2 + (4 + 3)x + 4 \times 3$$

$$= x^2 + 4x + 3x + 4 \times 3$$

$$= (x + 4)(x + 3)$$

Reason (R) : To factorise a given trinomial, the product of the first and the last term of the trinomial is always the sum of the two parts when we split the middle term.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

For a quadratic trinomial of the form :

$$ax^2 + bx + c$$

To factor using the middle-term splitting method;

Find two numbers that multiply to give ac and add to give b .

So, reason (R) is true.

$$\text{Given; } x^2 + 7x + 12$$

$$= x^2 + (3 + 4)x + 4 \times 3$$

$$= x^2 + 3x + 4x + 4 \times 3$$

$$= x(x + 3) + 4(x + 3)$$

$$= (x + 3)(x + 4)$$

So, assertion (A) is true and, reason (R) is the correct explanation of assertion (A).

Hence, option 1 is the correct option.

Question 1(ix)

Assertion (A) : The value of k so that the factors of $\left(x^2 - kx + \frac{121}{16}\right)$ the same is $\frac{11}{2}$.

Reason (R) : $(x + a)(x + b) = x^2 + (a + b)x + ab$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given; $\left(x^2 - kx + \frac{121}{16}\right)$

We know that this quadratic has equal factors if it's a perfect square trinomial, meaning :

$$\Rightarrow \left(x^2 - kx + \frac{121}{16}\right) = \left(x - \frac{11}{4}\right)^2$$

$$\Rightarrow \left(x^2 - kx + \frac{121}{16}\right) = x^2 - 2 \times x \times \frac{11}{4} + \left(\frac{11}{4}\right)^2$$

$$\Rightarrow \left(x^2 - kx + \frac{121}{16} \right) = x^2 - \frac{11}{2}x + \frac{121}{16}$$

$$\Rightarrow k = \frac{11}{2}.$$

So, assertion (A) is true.

Solving,

$$(x + a)(x + b) = x(x + b) + a(x + b)$$

$$= x^2 + bx + ax + ab$$

$$= x^2 + x(a + b) + ab$$

So, reason (R) is true but, reason (R) is not the correct explanation of assertion (A).

Hence, option 2 is the correct option.

Question 1(x)

Assertion (A) : There are two values of b so that $x^2 + bx - 24$ is factorisable.

Reason (R) : Two values have:

Product = -24 and sum = 2.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.

4. A is false, but R is true.

Answer

Given :

$$x^2 + bx - 24$$

To factor such a quadratic, we look for two numbers that :

Product = -24 and Sum = b

These values can be 6 and -4, -6 and 4, 1 and -24, -1 and 24, -3 and 8, -8 and 3, 12 and -2, 2 and -12.

In these cases value of b can be : $6 + (-4) = 2$, $(-6) + 4 = -2$, $1 + (-24) = -23$, $(-1) + 24 = 23$, $(-3) + 8 = 5$, $(-8) + 3 = -5$, $(-12) + 2 = -10$, $12 + (-2) = 10$.

Thus, there are more than 2 values of b for $x^2 + bx - 24$ to be factorisable.

∴ Assertion (A) is false.

Values have :

Product = -24 and sum = 2 or -2, 23 or -23, 10 or -10, 5 or -5.

∴ Reason (R) is false.

Hence, Both Assertion (A) and Reason (R) are false.

Question 2(i)

Factorise :

$$6x^3 - 8x^2$$

Answer

$$6x^3 - 8x^2$$

$$= 2x^2 (3x - 4)$$

$$\text{Hence, } 6x^3 - 8x^2 = 2x^2 (3x - 4)$$

Question 2(ii)

Factorise :

$$36x^2y^2 - 30x^3y^3 + 48x^3y^2$$

Answer

$$36x^2y^2 - 30x^3y^3 + 48x^3y^2$$

$$= 6x^2y^2 (6 - 5xy + 8x)$$

$$\text{Hence, } 36x^2y^2 - 30x^3y^3 + 48x^3y^2 = 6x^2y^2 (6 - 5xy + 8x)$$

Question 2(iii)

Factorise :

$$8(2a + 3b)^3 - 12(2a + 3b)^2$$

Answer

$$8(2a + 3b)^3 - 12(2a + 3b)^2$$

$$= 4(2(2a + 3b)^3 - 3(2a + 3b)^2)$$

$$= 4(2a + 3b)^2(2(2a + 3b) - 3)$$

$$= 4(2a + 3b)^2(4a + 6b - 3)$$

$$\text{Hence, } 8(2a + 3b)^3 - 12(2a + 3b)^2 = 4(2a + 3b)^2(4a + 6b - 3)$$

Question 2(iv)

Factorise :

$$9a(x - 2y)^4 - 12a(x - 2y)^3$$

Answer

$$9a(x - 2y)^4 - 12a(x - 2y)^3$$

$$= 3a(3(x - 2y)^4 - 4(x - 2y)^3)$$

$$= 3a(x - 2y)^3(3(x - 2y) - 4)$$

$$= 3a(x - 2y)^3(3x - 6y - 4)$$

$$\text{Hence, } 9a(x - 2y)^4 - 12a(x - 2y)^3 = 3a(x - 2y)^3(3x - 6y - 4)$$

Question 3(i)

Factorise :

$$a^2 - ab(1 - b) - b^3$$

Answer

$$a^2 - ab(1 - b) - b^3$$

$$= a^2 - ab + ab^2 - b^3$$

$$= (a^2 - ab) + (ab^2 - b^3)$$

$$= a(a - b) + b^2(a - b)$$

$$= (a - b)(a + b^2)$$

$$\text{Hence, } a^2 - ab(1 - b) - b^3 = (a - b)(a + b^2)$$

Question 3(ii)

Factorise :

$$xy^2 + (x - 1)y - 1$$

Answer

$$xy^2 + (x - 1)y - 1$$

$$= xy^2 + xy - y - 1$$

$$= (xy^2 + xy) - (y + 1)$$

$$= xy(y + 1) - 1(y + 1)$$

$$= (y + 1)(xy - 1)$$

$$\text{Hence, } xy^2 + (x - 1)y - 1 = (y + 1)(xy - 1)$$

Question 3(iii)

Factorise :

$$(ax + by)^2 + (bx - ay)^2$$

Answer

$$(ax + by)^2 + (bx - ay)^2$$

Using the formula

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$= a^2x^2 + b^2y^2 + 2abxy + b^2x^2 + a^2y^2 - 2abxy$$

$$= (a^2x^2 + a^2y^2) + (b^2y^2 + b^2x^2) + 2abxy - 2abxy$$

$$= a^2(x^2 + y^2) + b^2(x^2 + y^2)$$

$$= (x^2 + y^2)(a^2 + b^2)$$

$$\text{Hence, } (ax + by)^2 + (bx - ay)^2 = (x^2 + y^2)(a^2 + b^2)$$

Question 3(iv)

Factorise :

$$ab(x^2 + y^2) - xy(a^2 + b^2)$$

Answer

$$ab(x^2 + y^2) - xy(a^2 + b^2)$$

$$= abx^2 + aby^2 - xya^2 - xyb^2$$

$$= (abx^2 - xya^2) + (aby^2 - xyb^2)$$

$$= (abx^2 - a^2xy) + (aby^2 - b^2xy)$$

$$= ax(bx - ay) + by(ay - bx)$$

$$= ax(bx - ay) - by(bx - ay)$$

$$= (bx - ay)(ax - by)$$

$$\text{Hence, } ab(x^2 + y^2) - xy(a^2 + b^2) = (bx - ay)(ax - by)$$

Question 3(v)

Factorise :

$$m - 1 - (m - 1)^2 + am - a$$

Answer

$$m - 1 - (m - 1)^2 + am - a$$

$$= (m - 1) - (m - 1)^2 + (am - a)$$

$$= 1(m - 1) - (m - 1)^2 + a(m - 1)$$

$$= (m - 1)(1 - (m - 1) + a)$$

$$= (m - 1)(1 - m + 1 + a)$$

$$= (m - 1)(2 - m + a)$$

$$\text{Hence, } m - 1 - (m - 1)^2 + am - a = (m - 1)(2 - m + a)$$

Question 4(i)

Factorise :

$$25(2x - y)^2 - 16(x - 2y)^2$$

Answer

$$25(2x - y)^2 - 16(x - 2y)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (5(2x - y))^2 - (4(x - 2y))^2$$

$$= (5(2x - y) + 4(x - 2y))(5(2x - y) - 4(x - 2y))$$

$$= (10x - 5y + 4x - 8y)(10x - 5y - 4x + 8y)$$

$$= (10x + 4x - 5y - 8y)(10x - 4x - 5y + 8y)$$

$$= (14x - 13y)(6x + 3y)$$

$$= (14x - 13y)3(2x + y)$$

$$\text{Hence, } 25(2x - y)^2 - 16(x - 2y)^2 = 3(14x - 13y)(2x + y)$$

Question 4(ii)

Factorise :

$$16(5x + 4)^2 - 9(3x - 2)^2$$

Answer

$$16(5x + 4)^2 - 9(3x - 2)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (4(5x + 4))^2 - (3(3x - 2))^2$$

$$= (4(5x + 4) + 3(3x - 2))(4(5x + 4) - 3(3x - 2))$$

$$= (20x + 16 + 9x - 6)(20x + 16 - 9x + 6)$$

$$= (20x + 9x + 16 - 6)(20x - 9x + 16 + 6)$$

$$= (29x + 10)(11x + 22y)$$

$$= (29x + 10)11(x + 2y)$$

$$\text{Hence, } 16(5x + 4)^2 - 9(3x - 2)^2 = 11(29x + 10)(x + 2y)$$

Question 4(iii)

Factorise :

$$25(x - 2y)^2 - 4$$

Answer

$$25(x - 2y)^2 - 4$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (5(x - 2y))^2 - (2)^2$$

$$= (5(x - 2y) - 2)(5(x - 2y) + 2)$$

$$= (5x - 10y - 2)(5x - 10y + 2)$$

$$\text{Hence, } 25(x - 2y)^2 - 4 = (5x - 10y - 2)(5x - 10y + 2)$$

Question 5(i)

Factorise :

$$a^2 - 23a + 42$$

Answer

$$a^2 - 23a + 42$$

$$= a^2 - (21 + 2)a + 42$$

$$= a^2 - 21a - 2a + 42$$

$$= (a^2 - 21a) - (2a - 42)$$

$$= a(a - 21) - 2(a - 21)$$

$$= (a - 21)(a - 2)$$

$$\text{Hence, } a^2 - 23a + 42 = (a - 21)(a - 2)$$

Question 5(ii)

Factorise :

$$a^2 - 23a - 108$$

Answer

$$a^2 - 23a - 108$$

$$= a^2 - 23a - 108$$

$$= a^2 - (27 - 4)a - 108$$

$$= a^2 - 27a + 4a - 108$$

$$= (a^2 - 27a) + (4a - 108)$$

$$= a(a - 27) + 4(a - 27)$$

$$= (a - 27)(a + 4)$$

$$\text{Hence, } a^2 - 23a - 108 = (a - 27)(a + 4)$$

Question 5(iii)

Factorise :

$$1 - 18x - 63x^2$$

Answer

$$1 - 18x - 63x^2$$

$$= 1 - (21 - 3)x - 63x^2$$

$$= 1 - 21x + 3x - 63x^2$$

$$= (1 - 21x) + (3x - 63x^2)$$

$$= 1(1 - 21x) + 3x(1 - 9x)$$

$$= (1 - 21x)(1 + 3x)$$

$$\text{Hence, } 1 - 18x - 63x^2 = (1 - 21x)(1 + 3x)$$

Question 5(iv)

Factorise :

$$5x^2 - 4xy - 12y^2$$

Answer

$$5x^2 - 4xy - 12y^2$$

$$= 5x^2 - (10 - 6)xy - 12y^2$$

$$= 5x^2 - 10xy + 6xy - 12y^2$$

$$= (5x^2 - 10xy) + (6xy - 12y^2)$$

$$= 5x(x - 2y) + 6y(x - 2y)$$

$$= (x - 2y)(5x + 6y)$$

$$\text{Hence, } 5x^2 - 4xy - 12y^2 = (x - 2y)(5x + 6y)$$

Question 5(v)

Factorise :

$$x(3x + 14) + 8$$

Answer

$$x(3x + 14) + 8$$

$$= 3x^2 + 14x + 8$$

$$= 3x^2 + (12 + 2)x + 8$$

$$= 3x^2 + 12x + 2x + 8$$

$$= (3x^2 + 12x) + (2x + 8)$$

$$= 3x(x + 4) + 2(x + 4)$$

$$= (x + 4)(3x + 2)$$

$$\text{Hence, } x(3x + 14) + 8 = (x + 4)(3x + 2)$$

Question 5(vi)

Factorise :

$$5 - 4x(1 + 3x)$$

Answer

$$5 - 4x(1 + 3x)$$

$$= 5 - 4x - 12x^2$$

$$= 5 - (10 - 6)x - 12x^2$$

$$= 5 - 10x + 6x - 12x^2$$

$$= (5 - 10x) + (6x - 12x^2)$$

$$= 5(1 - 2x) + 6x(1 - 2x)$$

$$= (1 - 2x)(5 + 6x)$$

$$\text{Hence, } 5 - 4x(1 + 3x) = (1 - 2x)(5 + 6x)$$

Question 5(vii)

Factorise :

$$x^2y^2 - 3xy - 40$$

Answer

$$x^2y^2 - 3xy - 40$$

$$= x^2y^2 - (8 - 5)xy - 40$$

$$= x^2y^2 - 8xy + 5xy - 40$$

$$= (x^2y^2 - 8xy) + (5xy - 40)$$

$$= xy(xy - 8) + 5(xy - 8)$$

$$= (xy - 8)(xy + 5)$$

$$\text{Hence, } x^2y^2 - 3xy - 40 = (xy - 8)(xy + 5)$$

Question 5(viii)

Factorise :

$$(3x - 2y)^2 - 5(3x - 2y) - 24$$

Answer

$$(3x - 2y)^2 - 5(3x - 2y) - 24$$

$$= (3x - 2y)^2 - (8 - 3)(3x - 2y) - 24$$

$$= (3x - 2y)^2 - 8(3x - 2y) + 3(3x - 2y) - 24$$

$$= (3x - 2y)((3x - 2y) - 8) + 3((3x - 2y) - 8)$$

$$= (3x - 2y)(3x - 2y - 8) + 3(3x - 2y - 8)$$

$$= (3x - 2y - 8)(3x - 2y + 3)$$

$$\text{Hence, } (3x - 2y)^2 - 5(3x - 2y) - 24 = (3x - 2y - 8)(3x - 2y + 3)$$

Question 5(ix)

Factorise :

$$12(a + b)^2 - (a + b) - 35$$

Answer

$$12(a + b)^2 - (a + b) - 35$$

$$= 12(a + b)^2 - (21 - 20)(a + b) - 35$$

$$= 12(a + b)^2 - 21(a + b) + 20(a + b) - 35$$

$$= 3(a + b)(4(a + b) - 7) + 5(4(a + b) - 7)$$

$$= 3(a + b)(4a + 4b - 7) + 5(4a + 4b - 7)$$

$$= (4a + 4b - 7)(3(a + b) + 5)$$

$$= (4a + 4b - 7)(3a + 3b + 5)$$

$$\text{Hence, } 12(a + b)^2 - (a + b) - 35 = (4a + 4b - 7)(3a + 3b + 5)$$

Question 6(i)

Factorise :

$$15(5x - 4)^2 - 10(5x - 4)$$

Answer

$$15(5x - 4)^2 - 10(5x - 4)$$

$$= 5(5x - 4)(3(5x - 4) - 2)$$

$$= 5(5x - 4)(15x - 12 - 2)$$

$$= 5(5x - 4)(15x - 14)$$

$$\text{Hence, } 15(5x - 4)^2 - 10(5x - 4) = 5(5x - 4)(15x - 14)$$

Question 6(ii)

Factorise :

$$3a^2x - bx + 3a^2 - b$$

Answer

$$3a^2x - bx + 3a^2 - b$$

$$= x(3a^2 - b) + 1(3a^2 - b)$$

$$= (3a^2 - b)(x + 1)$$

$$\text{Hence, } 3a^2x - bx + 3a^2 - b = (3a^2 - b)(x + 1)$$

Question 6(iii)

Factorise :

$$b(c - d)^2 + a(d - c) + 3(c - d)$$

Answer

$$b(c - d)^2 + a(d - c) + 3(c - d)$$

$$= b(c - d)^2 - a(c - d) + 3(c - d)$$

$$= (c - d)(b(c - d) - a + 3)$$

$$= (c - d)(bc - bd - a + 3)$$

$$\text{Hence, } b(c - d)^2 + a(d - c) + 3(c - d) = (c - d)(bc - bd - a + 3)$$

Question 6(iv)

Factorise :

$$ax^2 + b^2y - ab^2 - x^2y$$

Answer

$$\begin{aligned} & ax^2 + b^2y - ab^2 - x^2y \\ &= (ax^2 - ab^2) + (b^2y - x^2y) \\ &= a(x^2 - b^2) + y(b^2 - x^2) \\ &= a(x^2 - b^2) - y(x^2 - b^2) \\ &= (x^2 - b^2)(a - y) \end{aligned}$$

Using the formula

$$\begin{aligned} & [\because (x^2 - y^2) = (x + y)(x - y)] \\ &= (x - b)(x + b)(a - y) \end{aligned}$$

$$\text{Hence, } ax^2 + b^2y - ab^2 - x^2y = (x - b)(x + b)(a - y)$$

Question 6(v)

Factorise :

$$1 - 3x - 3y - 4(x + y)^2$$

Answer

$$\begin{aligned} & 1 - 3x - 3y - 4(x + y)^2 \\ &= 1 - 3(x + y) - 4(x + y)^2 \\ &= 1 - (4 - 1)(x + y) - 4(x + y)^2 \\ &= 1 - 4(x + y) + 1(x + y) - 4(x + y)^2 \end{aligned}$$

$$= (1 - 4(x + y)) + (1(x + y) - 4(x + y)^2)$$

$$= 1(1 - 4(x + y)) + (x + y)(1 - 4(x + y))$$

$$= 1(1 - 4x - 4y) + (x + y)(1 - 4x - 4y)$$

$$= (1 - 4x - 4y)(1 + (x + y))$$

$$= (1 - 4x - 4y)(1 + x + y)$$

$$\text{Hence, } 1 - 3x - 3y - 4(x + y)^2 = (1 - 4x - 4y)(1 + x + y)$$

Question 7(i)

Factorise :

$$2a^3 - 50a$$

Answer

$$2a^3 - 50a$$

$$= 2a(a^2 - 25)$$

$$= 2a(a^2 - 5^2)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= 2a(a - 5)(a + 5)$$

$$\text{Hence, } 2a^3 - 50a = 2a(a - 5)(a + 5)$$

Question 7(ii)

Factorise :

$$54a^2b^2 - 6$$

Answer

$$54a^2b^2 - 6$$

$$= 6(9a^2b^2 - 1)$$

$$= 6((3ab)^2 - (1)^2)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= 6(3ab - 1)(3ab + 1)$$

$$\text{Hence, } 54a^2b^2 - 6 = 6(3ab - 1)(3ab + 1)$$

Question 7(iii)

Factorise :

$$64a^2b - 144b^3$$

Answer

$$64a^2b - 144b^3$$

$$= 16b(4a^2 - 9b^2)$$

$$= 16b((2a)^2 - (3b)^2)$$

$$= 16b(2a + 3b)(2a - 3b)$$

$$\text{Hence, } 64a^2b - 144b^3 = 16b(2a + 3b)(2a - 3b)$$

Question 7(iv)

Factorise :

$$(2x - y)^3 - (2x - y)$$

Answer

$$(2x - y)^3 - (2x - y)$$

$$= (2x - y)((2x - y)^2 - 1)$$

$$= (2x - y)((2x - y)^2 - (1)^2)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (2x - y)((2x - y) - 1)((2x - y) + 1)$$

$$= (2x - y)(2x - y - 1)(2x - y + 1)$$

$$\text{Hence, } (2x - y)^3 - (2x - y) = (2x - y)(2x - y - 1)(2x - y + 1)$$

Question 7(v)

Factorise :

$$x^2 - 2xy + y^2 - z^2$$

Answer

$$x^2 - 2xy + y^2 - z^2$$

Using the formula

$$[\because (x - y)^2 = x^2 + y^2 - 2xy]$$

$$= (x^2 - 2xy + y^2) - z^2$$

$$= (x - y)^2 - z^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((x - y) - z)((x - y) + z)$$

$$= (x - y - z)(x - y + z)$$

$$\text{Hence, } x^2 - 2xy + y^2 - z^2 = (x - y - z)(x - y + z)$$

Question 7(vi)

Factorise :

$$x^2 - y^2 - 2yz - z^2$$

Answer

$$x^2 - y^2 - 2yz - z^2$$

Using the formula

$$[\because (x - y)^2 = x^2 + y^2 - 2xy]$$

$$= x^2 - (y^2 + 2yz + z^2)$$

$$= x^2 - (y + z)^2$$

$$= (x)^2 - (y + z)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (x - (y + z))(x + (y + z))$$

$$= (x - y - z)(x + y + z)$$

$$\text{Hence, } x^2 - y^2 - 2yz - z^2 = (x - y - z)(x + y + z)$$

Question 7(vii)

Factorise :

$$7a^5 - 567a$$

Answer

$$7a^5 - 567a$$

$$= 7a(a^4 - 81)$$

$$= 7a((a^2)^4 - (9)^4)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= 7a(a^2 - 9)(a^2 + 9)$$

$$= 7a((a)^2 - (3)^2)(a^2 + 9)$$

$$= 7a(a - 3)(a + 3)(a^2 + 9)$$

$$\text{Hence, } 7a^5 - 567a = 7a(a - 3)(a + 3)(a^2 + 9)$$

Question 7(viii)

Factorise :

$$5x^2 - \frac{20x^4}{9}$$

Answer

$$5x^2 - \frac{20x^4}{9}$$

$$= 5x^2 \left(1 - \frac{4x^2}{9} \right)$$

$$= 5x^2 \left(1^2 - \left(\frac{2x}{3} \right)^2 \right)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= 5x^2 \left(1 - \frac{2x}{3}\right) \left(1 + \frac{2x}{3}\right)$$

$$\text{Hence, } 5x^2 - \frac{20x^4}{9} = 5x^2 \left(1 - \frac{2x}{3}\right) \left(1 + \frac{2x}{3}\right)$$

Question 8

Factorise $xy^2 - xz^2$, Hence, find the value of :

$$(i) 9 \times 8^2 - 9 \times 2^2$$

$$(ii) 40 \times 5.5^2 - 40 \times 4.5^2$$

Answer

$$xy^2 - xz^2$$

$$= x(y^2 - z^2)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= x(y - z)(y + z)$$

$$\text{Hence, } xy^2 - xz^2 = x(y - z)(y + z)$$

$$(i) 9 \times 8^2 - 9 \times 2^2$$

$$= 9 \times (8 - 2)(8 + 2)$$

$$= 9 \times 6 \times 10$$

$$= 540$$

$$\text{Hence, } 9 \times 8^2 - 9 \times 2^2 = 540$$

$$(ii) 40 \times 5.5^2 - 40 \times 4.5^2$$

$$= 40 \times (5.5 - 4.5) \times (5.5 + 4.5)$$

$$= 40 \times 1 \times 10$$

$$= 400$$

$$\text{Hence, } 40 \times 5.5^2 - 40 \times 4.5^2 = 400$$

Question 9(i)

Factorise :

$$(a - 3b)^2 - 36 b^2$$

Answer

$$(a - 3b)^2 - 36 b^2$$

$$= (a - 3b)^2 - (6 b)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= ((a - 3b) - 6b)((a - 3b) + 6b)$$

$$= (a - 3b - 6b)(a - 3b + 6b)$$

$$= (a - 9b)(a + 3b)$$

$$\text{Hence, } (a - 3b)^2 - 36b^2 = (a - 9b)(a + 3b)$$

Question 9(ii)

Factorise :

$$25(a - 5b)^2 - 4(a - 3b)^2$$

Answer

$$25(a - 5b)^2 - 4(a - 3b)^2$$

$$= (5(a - 5b))^2 - (2(a - 3b))^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (5(a - 5b) - 2(a - 3b))((5(a - 5b)) + 2(a - 3b))$$

$$= (5a - 25b - 2a + 6b)(5a - 25b + 2a - 6b)$$

$$= (5a - 2a - 25b + 6b)(5a + 2a - 25b - 6b)$$

$$= (3a - 19b)(7a - 31b)$$

$$\text{Hence, } 25(a - 5b)^2 - 4(a - 3b)^2 = (3a - 19b)(7a - 31b)$$

Question 9(iii)

Factorise :

$$a^2 - 0.36b^2$$

Answer

$$a^2 - 0.36 b^2$$

$$= (a)^2 - (0.6 b)^2$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= (a + 0.6b)(a - 0.6b)$$

$$\text{Hence, } a^2 - 0.36 b^2 = (a + 0.6b)(a - 0.6b)$$

Question 9(iv)

Factorise :

$$x^4 - 5x^2 - 36$$

Answer

$$x^4 - 5x^2 - 36$$

$$= x^4 - (9 - 4)x^2 - 36$$

$$= x^4 - 9x^2 + 4x^2 - 36$$

$$= (x^4 - 9x^2) + (4x^2 - 36)$$

$$= x^2(x^2 - 9) + 4(x^2 - 9)$$

$$= (x^2 - 9)(x^2 + 4)$$

$$= ((x)^2 - (3)^2)(x^2 + 4)$$

$$= (x - 3)(x + 3)(x^2 + 4)$$

$$\text{Hence, } x^4 - 5x^2 - 36 = (x - 3)(x + 3)(x^2 + 4)$$

Question 9(v)

Factorise :

$$15(2x - y)^2 - 16(2x - y) - 15$$

Answer

$$15(2x - y)^2 - 16(2x - y) - 15$$

$$= 15(2x - y)^2 - (25 - 9)(2x - y) - 15$$

$$= 15(2x - y)^2 - 25(2x - y) + 9(2x - y) - 15$$

$$= (15(2x - y)^2 - 25(2x - y)) + (9(2x - y) - 15)$$

$$= 5(2x - y)(3(2x - y) - 5) + 3(3(2x - y) - 5)$$

$$= 10x - 5y(6x - 3y - 5) + 3(6x - 3y - 5)$$

$$= (6x - 3y - 5)(10x - 5y + 3)$$

$$\text{Hence, } 15(2x - y)^2 - 16(2x - y) - 15 = (6x - 3y - 5)(10x - 5y + 3)$$

Question 10

Evaluate (using factors) : $301^2 \times 300 - 300^3$

Answer

$$301^2 \times 300 - 300^3$$

$$= 300 \times (301^2 - 300^2)$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$= 300 \times (301 - 300) \times (301 + 300)$$

$$= 300 \times 1 \times 601$$

$$= 1,80,300$$

$$\text{Hence, } 301^2 \times 300 - 300^3 = 1,80,300$$

Question 11(i)

Use factor method to evaluate:

$$(5z^2 - 80) \div (z - 4)$$

Answer

$$\frac{(5z^2 - 80)}{(z - 4)}$$

$$= \frac{5(z^2 - 16)}{z - 4}$$

Using the formula

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$\frac{5(z^2 - 4^2)}{(z - 4)}$$

$$= \frac{5(z - 4)(z + 4)}{(z - 4)}$$

$$= \frac{5\cancel{(z - 4)}(z + 4)}{\cancel{(z - 4)}}$$

$$= 5(z + 4)$$

$$\text{Hence, } (5z^2 - 80) \div (z - 4) = 5(z + 4)$$

Question 11(ii)

Use factor method to evaluate:

$$10y(6y + 21) \div (2y + 7)$$

Answer

$$\frac{10y(6y + 21)}{(2y + 7)}$$

$$\frac{10y \times 3(2y + 7)}{(2y + 7)}$$

$$= \frac{30y(2y + 7)}{2y + 7}$$

$$= \frac{30y\cancel{(2y+7)}}{\cancel{(2y+7)}}$$

$$= 30y$$

$$\text{Hence, } 10y(6y + 21) \div (2y + 7) = 30y$$

Question 11(iii)

Use factor method to evaluate:

$$(a^2 - 14a - 32) \div (a + 2)$$

Answer

$$\begin{aligned} & \frac{(a^2 - 14a - 32)}{(a + 2)} \\ &= \frac{(a^2 - (16 - 2)a - 32)}{(a + 2)} \\ &= \frac{(a^2 - 16a + 2a - 32)}{(a + 2)} \\ &= \frac{((a^2 - 16a) + (2a - 32))}{(a + 2)} \\ &= \frac{(a(a - 16) + 2(a - 16))}{(a + 2)} \\ &= \frac{((a - 16)(a + 2))}{(a + 2)} \end{aligned}$$

$$= \frac{(a - 16)\cancel{(a + 2)}}{\cancel{(a + 2)}}$$

$$= (a - 16)$$

$$\text{Hence, } (a^2 - 14a - 32) \div (a + 2) = (a - 16)$$

Question 11(iv)

Use factor method to evaluate:

$$39x^3(50x^2 - 98) \div 26x^2(5x + 7)$$

Answer

$$\frac{39x^3(50x^2 - 98)}{26x^2(5x + 7)}$$

$$= \frac{3x(50x^2 - 98)}{2(5x + 7)}$$

$$= \frac{3x \times 2(25x^2 - 49)}{2(5x + 7)}$$

$$= \frac{6x((5x)^2 - (7)^2)}{2(5x + 7)}$$

$$= \frac{3x(5x - 7)(5x + 7)}{(5x + 7)}$$

$$= \frac{3x(5x - 7)\cancel{(5x + 7)}}{\cancel{(5x + 7)}}$$

$$= 3x(5x - 7)$$

$$\text{Hence, } 39x^3(50x^2 - 98) \div 26x^2(5x + 7) = 3x(5x - 7)$$

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